

Numerical Algorithms

Problem set 5: Basic linear algebra

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[**Note:**] If the numerical matrix computations turn out to be tedious, you may use the function `Matrix().rref` in the Python `sympy` package.

1 Row-reduced echelon matrices

1. Consider the following systems of equation

(a)

$$\begin{array}{rrrrrrcl} x_1 & - & x_2 & + & 2x_3 & = & 1 \\ 2x_1 & & & + & 2x_3 & = & 1 \\ x_1 & - & 3x_2 & + & 4x_3 & = & 2 \end{array}$$

(b)

$$\begin{array}{rrrrrrcl} x_1 & - & 2x_2 & + & x_3 & + & 2x_4 & = & 1 \\ x_1 & + & x_2 & - & x_3 & + & x_4 & = & 2 \\ x_1 & + & 7x_2 & - & 5x_3 & - & x_4 & = & 3 \end{array}$$

Find out whether they have solutions. If so, describe explicitly all solutions.

2. Let

$$\mathbf{A} = \begin{bmatrix} 3 & -1 & 2 \\ 2 & 1 & 1 \\ 1 & -3 & 0 \end{bmatrix}.$$

for which triples (y_1, y_2, y_3) does the system $\mathbf{A}\mathbf{X} = \mathbf{Y}$ have a solution?

3. Let

$$\mathbf{A} = \begin{bmatrix} 3 & -6 & 2 & -1 \\ -2 & 4 & 1 & 3 \\ 0 & 0 & 1 & 1 \\ 1 & -2 & 1 & 0 \end{bmatrix}$$

for which (y_1, y_2, y_3, y_4) does the system $\mathbf{AX} = \mathbf{Y}$ have a solution?

4. Suppose $\mathbf{R}$ and $\mathbf{R}'$ are 2×3 row-reduced echelon matrices and that the systems $\mathbf{RX} = \mathbf{0}$ and $\mathbf{R}'\mathbf{X} = \mathbf{0}$ have exactly the same solutions. Prove that $\mathbf{R} = \mathbf{R}'$

5. Let

$$\mathbf{A} = \begin{bmatrix} 1 & 2 & 1 & 0 \\ -1 & 0 & 3 & 5 \\ 1 & -2 & 1 & 1 \end{bmatrix}$$

Find a row-reduced echelon matrix $\mathbf{R}$ which is row-equivalent to $\mathbf{A}$ and an invertible 3×3 matrix $\mathbf{P}$ such that $\mathbf{R} = \mathbf{PA}$.

6. For each of the three matrices

$$\begin{bmatrix} 2 & 5 & -1 \\ 4 & -1 & 2 \\ 6 & 4 & 1 \end{bmatrix}, \begin{bmatrix} 1 & -1 & 2 \\ 3 & 2 & 4 \\ 0 & 1 & -2 \end{bmatrix}, \begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 2 & 3 & 4 \\ 0 & 0 & 3 & 4 \\ 0 & 0 & 0 & 4 \end{bmatrix}$$

use elementary row operations to discover whether it is invertible, and to find the inverse in case it is.

7. Suppose $\mathbf{A}$ is a 2×1 matrix and that $\mathbf{B}$ is a 1×2 matrix. Prove that $\mathbf{C} = \mathbf{AB}$ is not invertible.

8. Let $\mathbf{A}$ be an $n \times n$ matrix. Prove the following:

- (a) If $\mathbf{A}$ is invertible and $\mathbf{AB} = \mathbf{0}$ for some $n \times n$ matrix $\mathbf{B}$, the $\mathbf{B} = \mathbf{0}$.
- (b) If $\mathbf{A}$ is not invertible, then there exists an $n \times n$ matrix $\mathbf{B}$ such that $\mathbf{AB} = \mathbf{0}$ but $\mathbf{B} \neq \mathbf{0}$.

9. Let

$$\mathbf{A} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

Prove, using elementary row operations, that $\mathbf{A}$ is invertible *if and only if* $(ad - bc) \neq 0$.

10. Let

$$\mathbf{A} = \begin{bmatrix} 5 & 0 & 0 \\ 1 & 5 & 0 \\ 0 & 1 & 5 \end{bmatrix}$$

For which $\mathbf{X}$ does there exist a scalar c such that $\mathbf{AX} = c\mathbf{X}$?

11. An $n \times n$ matrix $\mathbf{A}$ is **upper-triangular** if $A_{ij} = 0$ for $i > j$. Prove that $\mathbf{A}$ is invertible *if and only if* every entry on its main diagonal is distinct from 0.
12. Prove that if $\mathbf{A}$ is an $m \times n$ matrix, $\mathbf{B}$ is an $n \times m$ matrix and $n < m$, then $\mathbf{AB}$ is not invertible (generalization of a previous problem).
13. Let $\mathbf{A}$ be an $m \times n$ matrix. Show that by means of a finite number of elementary row and/or column operations one can pass from $\mathbf{A}$ to a matrix $\mathbf{R}$ which is both *row-reduced echelon* and *column-reduced echelon*, i.e., $R_{ij} = 0$ if $i \neq j$, $R_{ii} = 1, 1 \leq i \leq r$, $R_{ii} = 0, i > r$. Show that $\mathbf{R} = \mathbf{PAQ}$ where $\mathbf{P}$ is an invertible $n \times n$ matrix and $\mathbf{Q}$ is an invertible $m \times m$ matrix.

2 Vector spaces and subspaces

1. Show that the following are vector spaces:
 - (a) The n -tuple space, F^n : Let F be a Field and let V be the set of all n -tuples $\alpha = (x_1, x_2, \dots, x_n)$ of scalars $x_i \in F$. If $\beta = (y_1, y_2, \dots, y_n)$ with $y_i \in F$ then their sum is defined as

$$\alpha + \beta = (x_1 + y_1, \dots, x_n + y_n)$$

and the product of a scalar c and a vector α is

$$c\alpha = (cx_1, cx_2, \dots, cx_n)$$

- (b) The space of $m \times n$ matrices, $F^{m \times n}$: under usual matrix addition and multiplication of a matrix with a scalar.
 - (c) The space of functions from a set to a Field: under the operations:

$$(f + g)(s) = f(s) + g(s)$$

and

$$(cf)(s) = cf(s)$$

- (d) The space of polynomial functions over a Field: with addition and scalar multiplication as defined above.
2. Which of the following sets of vectors $\alpha = (a_1, \dots, a_n)$ in $\mathbb{R}^n$ are subspaces of $\mathbb{R}^n$ ($n \geq 3$)?

- (a) all α such that $a_1 \geq 0$;
 - (b) all α such that $a_1 + 3a_2 = a_3$;
 - (c) all α such that $a_2 = a_1^2$;
 - (d) all α such that $a_1 a_2 = 0$;
 - (e) all α such that a_2 is rational.
3. Let V be the (real) vector space of all functions $f : \mathbb{R} \rightarrow \mathbb{R}$. Which of the following are subspaces of V ?
- (a) all f such that $f(x^2) = f(x)^2$;
 - (b) all f such that $f(0) = f(1)$;
 - (c) all f such that $f(3) = 1 + f(-5)$;
 - (d) all f such that $f(-1) = 0$;
 - (e) all f which are continuous.
4. Let W be the set of all $(x_1, x_2, x_3, x_4, x_5) \in \mathbb{R}^5$ which satisfy

$$\begin{array}{ccccccccc} 2x_1 & - & x_2 & + & \frac{4}{3}x_3 & - & x_4 & & = & 0 \\ x_1 & & & + & \frac{2}{3}x_3 & & & - & x_5 & = & 0 \\ 9x_1 & - & 3x_2 & + & 6x_3 & - & 3x_4 & - & 3x_5 & = & 0 \end{array}$$

Find a finite set of vectors which spans W .

5. Let F be a Field and let n be a positive integer ($n \geq 2$). Let V be a vector space of all $n \times n$ matrices over F . Which of the following set of matrices A in V are subspaces of V ?
- (a) all invertible A ;
 - (b) all non-invertible A ;
 - (c) all A such that $AB = BA$, where B is some fixed matrix in V ;
 - (d) all A such that $A^2 = A$.
6. (a) Prove that the only subspaces of $\mathbb{R}^1$ are $\mathbb{R}^1$ and the zero subspace.
 (b) Prove that a subspace of $\mathbb{R}^2$ is $\mathbb{R}^2$, or the zero subspace, or consists of all scalar multiples of some fixed vector in $\mathbb{R}^2$.
 (c) What can you say about the subspaces of $\mathbb{R}^3$?
7. Let $\mathbf{A}$ be a $m \times n$ matrix over F . Show that the set of all vectors $\mathbf{X}$ such that $\mathbf{AX} = \mathbf{0}$ is a subspace of F^n . This subspace is called the **null space** of $\mathbf{A}$ and its dimension is the **nullity** of $\mathbf{A}$.

8. Let $\mathbf{A}$ be a $m \times n$ matrix over F . Show that the set of all vectors spanned by the row vectors of $\mathbf{A}$ is a subspace of F^n . This subspace is called the **row space** of $\mathbf{A}$ and its dimension is the **row rank** of $\mathbf{A}$.
9. Let $\mathbf{A}$ be a $m \times n$ matrix over F . Show that the set of all vectors $\mathbf{Y}$ such that $\mathbf{AX} = \mathbf{Y}$ has a solution for $\mathbf{X}$ is a subspace of F^m . This subspace is called the **range space** of $\mathbf{A}$ and its dimension is the **column rank** of $\mathbf{A}$ (why?).
10. Show that for any matrix $\mathbf{A}$
 - (a) **nullity** + **row rank** = n
 - (b) **nullity** + **column rank** = n

and conclude that

row rank of $\mathbf{A}$ = **column rank** of $\mathbf{A}$

11. Consider the 5×5 matrix

$$\mathbf{A} = \begin{bmatrix} 1 & 2 & 0 & 3 & 0 \\ 1 & 2 & -1 & -1 & 0 \\ 0 & 0 & 1 & 4 & 0 \\ 2 & 4 & 1 & 10 & 1 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

- (a) Find an invertible matrix $\mathbf{P}$ such that $\mathbf{PA}$ is a row-reduced echelon matrix $\mathbf{R}$.
- (b) Find a basis for the **row space** W of $\mathbf{R}$.
- (c) Say which vectors $(b_1, b_2, b_3, b_4, b_5)$ are in W .
- (d) Find the coordinate matrix of each vector $(b_1, b_2, b_3, b_4, b_5) \in W$ in the ordered basis chosen in (b).
- (e) Write each vector $(b_1, b_2, b_3, b_4, b_5) \in W$ as a linear combination of the rows of $\mathbf{A}$.
- (f) Give an explicit description of the **null space** of $\mathbf{A}$.
- (g) Find a basis for the **null space**.
- (h) For what column matrices $\mathbf{Y}$ does the equation $\mathbf{AX} = \mathbf{Y}$ have solutions $\mathbf{X}$?
- (i) Explicitly find the **range space** of $\mathbf{A}$ and find a basis.
- (j) Verify the relations regarding the **nullity**, **row rank** and **column rank** of $\mathbf{A}$.

3 Coordinates

1. Let $\mathcal{B} = \{\alpha_1, \alpha_2, \alpha_3\}$ be an ordered basis for $\mathbb{R}^3$ where

$$\alpha_1 = (1, 0, -1), \quad \alpha_2 = (1, 1, 1), \quad \alpha_3 = (1, 0, 0)$$

What are the coordinates of (a, b, c) in the ordered basis $\mathcal{B}$?

2. Let $\alpha = (x_1, x_2)$ and $\beta = (y_1, y_2)$ be vectors in $\mathbb{R}^2$ such that

$$x_1 y_1 + x_2 y_2 = 0 \text{ and } x_1^2 + x_2^2 = y_1^2 + y_2^2 = 1$$

Show that $\mathcal{B} = \{\alpha, \beta\}$ is a basis for $\mathbb{R}^2$. Find the coordinates of (a, b) in this ordered basis. What do the conditions mean geometrically?

3. Consider the matrix

$$\mathbf{P} = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$$

Show that $\mathbf{P}$ is invertible. Conclude that $\mathbf{P}$ represents a transformation of coordinates in $\mathbb{R}^2$. What is the geometric interpretation of the change of coordinates represented by $\mathbf{P}$?

4. Let V be a vector space over the complex numbers of all functions from $\mathbb{R}$ to $\mathbb{C}$, i.e., the space of all complex valued functions on the real line. Let $f_1(x) = 1$, $f_2(x) = e^{ix}$ and $f_3(x) = e^{-ix}$.

- (a) Prove that f_1, f_2, f_3 are linearly independent.
(b) Let $g_1(x) = 1$, $g_2(x) = \cos x$ and $g_3(x) = \sin x$. Find an invertible 3×3 matrix $\mathbf{P}$ such that

$$g_j = \sum_{i=1}^3 P_{ij} f_i$$

5. Let V be the real vector space of all polynomial functions from $\mathbb{R}$ into $\mathbb{R}$ of degree 2 or less. Let t be a fixed number and define

$$g_1(x) = 1, \quad g_2(x) = x + t, \quad g_3(x) = (x + t)^2$$

Prove that $\mathcal{B} = \{g_1, g_2, g_3\}$ is a basis for V . If

$$f(x) = c_0 + c_1 x + c_2 x^2$$

what are the coordinates of f in this ordered basis $\mathcal{B}$?